

Solution Requirements

Date	03 July 2026
Team ID	SWTID-2026-4350
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system maintains user accounts (Admin and Analyst) and verifies users before accessing application features.</i>	<i>High</i>
2	Authorization levels	<i>Admin manages the application and ML model information, while the Analyst performs loan prediction and manages applicant records.</i>	<i>High</i>
3	External interfaces	<i>The application uses the CreditPredictor module (src/prediction.py) to generate loan approval predictions.</i>	<i>High</i>

4	Transactions processing	<i>The system collects applicant details, stores them in the database, performs prediction, assigns a risk category, and saves the prediction result.</i>	<i>High</i>
5	Reporting	<i>The dashboard displays total applicants, total predictions, approval rate, rejection rate, active ML model, and recent prediction history.</i>	<i>Medium</i>
6	Business rules, etc.	<i>Loan approval is predicted using the Machine Learning model. Risk Category is assigned based on prediction confidence: Low Risk (≥ 0.80), Medium Risk (≥ 0.50), High Risk (< 0.50).</i>	<i>High</i>
7	Compliance to laws or regulations	<i>Applicant information should be securely stored and accessed only by authorized users.</i>	<i>Medium</i>
8	Other	<i>The application initializes the database automatically, creates required tables, and inserts default users and ML model information during startup.</i>	<i>Medium</i>

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>The application should generate loan predictions and display results without noticeable delay.</i>	<i>Prediction result displayed within 3 seconds.</i>
2	Scalability	<i>The application should efficiently manage increasing applicant records and prediction history.</i>	<i>Database should support thousands of applicant records.</i>
3	Security & Data Privacy	<i>Applicant information should be stored securely, and only authorized users should have access to application data.</i>	<i>Secure storage of applicant and prediction data.</i>
4	Reliability & Availability	<i>The application should initialize correctly, create required database tables if missing, and provide uninterrupted prediction functionality.</i>	<i>Application starts successfully and remains available during operation.</i>
5	Usability & Accessibility	<i>The web interface should be simple, intuitive, and easy to use for entering applicant details and viewing prediction results.</i>	<i>Users should be able to complete a prediction with minimal guidance.</i>
6	Other	<i>The project should have a modular structure (app.py, config.py, src/prediction.py, templates) to simplify maintenance and future updates.</i>	<i>Modular code structure with separate configuration, prediction, and presentation components.</i>